

PS2: CAD-to-Quote AI Agent

Instant, transparent manufacturing
quotes from 3D part data

TEAM: H@CTIVATOR\$

- PIVOT Innovation Challenge · Round 2
- Product Development Centre, L&T



What We've Built

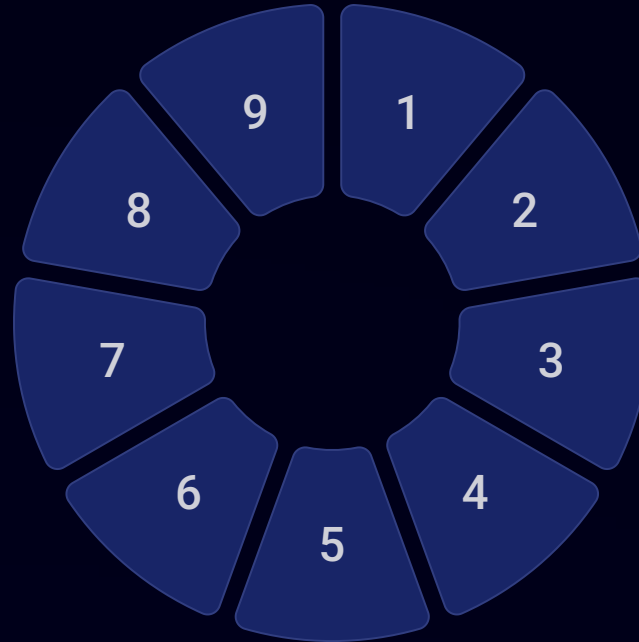
Implementation Status

Real STEP parsing (samples simulate output)

ML model training (placeholder active)

17-feature ML schema designed

Confidence intervals and risk classification



9 interactive frontend screens

FastAPI backend with modular architecture

Process routing engine with warnings

4-component cost calculation engine

DFM suggestion engine with re-prediction

Live Demo Walkthrough

Steel Bracket — 9 Screens

01

Landing

Select Steel Bracket sample

02

Geometry

Volume, area, bbox, faces

03

Input

6 holes · Steel · Qty 50 · Fine tolerance

04

Routing

Milling  · Drilling  · Grinding  ·
Turning 

05

Stock

Rectangular block · 2.04 kg

06

Progress

7-step AI pipeline animation

07

Dashboard

₹28,400 · Full breakdown · SHAP · DFM

08

DFM

Original vs. Optimized side-by-side

09

Export

PDF-ready quote summary

Round 1 → Round 2 Progress

What Changed

Round 1

- High-level concept
- No code
- No UI
- No DFM
- No explainability

Round 2

- 13-step pipeline architecture
- 9 backend modules designed
- Working web app (9 screens)
- Professional dark-theme dashboard
- DFM view with ₹ savings
- 3-layer explainability

Impact & KPIs

Expected Benefits

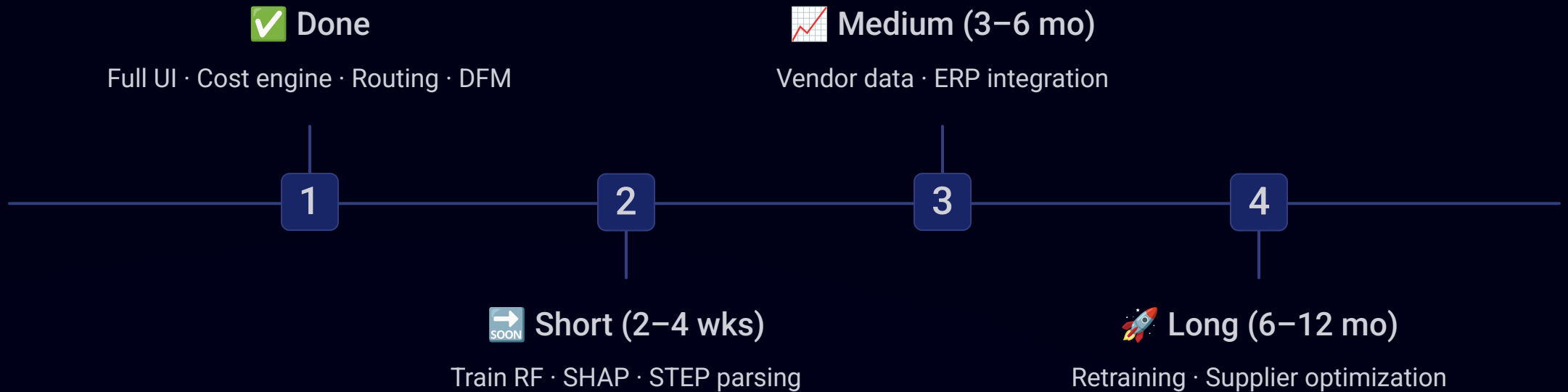
Metric	Before	After
Quote time	Hours–days	< 30 sec
Transparency	✗ None	✓ Full breakdown
DFM timing	Post-production	Design stage
Estimator need	Senior expert	Automated

Honest Limitations:

- ML currently uses formula placeholder
- STEP parsing simulated via samples
- Costs use representative rates (configurable)

Roadmap & Close

What's Next



From CAD geometry to a transparent, explainable quote — in seconds.

Thank you. We're ready for your questions.